

After we had completed Tuesday's objectives I was contemplating if we had missed a key factor for the project. this position that this project is for is the IT specialist AI roll for the US government. and for the first test was to generate a comparison report on the book The Great Gatsby translated to Spanish and then translated to Mandarin only using AI No handwritten anything. so this test I was thinking Although our project is well documented we didn't have any strong AI presence within our project. a project does use OCR to read the text on the labels and to normalize it but other than that we didn't really have an AI presence. after talking to Chad about it our plan is to continue working what we have documented and then later on if we have time think about adding mIn module into the project for better readouts.

after completing Thursday's objectives of getting the CSV files loading and processing I realize that some of the test labels that I've downloaded from the internet the OCR model was having a terrible time we can get and I was wondering the question that came up is why this chat should be tea have a better time understanding what a label is or what are images that I send it compared to the weed out I get from OCR using that same exact image. do that conversation I realize we may need to use a BLM visual language model until this project but again like as I said it before you will stay on task finish the plan that we worked out and once we have time to add something else to enhance our project.

It's Saturday and we are already working on Sundays activities this project has been semi-abreeze I think that's because we took a lot of time to dictate our project sculpt and exactly what we want to accomplish we would have been farther ahead I was off work Thursday however since we were ahead I took some time to get some rest since I'm back at a physical job for the intermediate time. today's objective was to create data to test our application and then to deploy it. in creating data I scraped some labels straight from the ttb website. as are controlled clean labels. and then we used Chad gpt's image generator to generate a couple of clean labels a 6 poor quality labels as in too much text on the label blurry text slanted text and so on. and then we used it to create six invalid labels which means labels that are missing the government warning or alcohol contents or net content so on and so forth. once those were generated I updated the master CSV to include each label on the lines we now have Clean valid with poor quality then we have invalid missing alcohol content invalid missing net content invalid missing warning. I took the file the CSV file and the images put them into our web application ran the test I noticed we have a few issues or the first issue on mine in the CSV file I had a couple of Records empty records they just had the record number no information so some of our poor quality or invalid labels were matching with those no records so I clean that up by deleting the record number just so that our file is most valid and accurate to read the main notice is that the labels that were supposed to and pass weren't passing due to the exact matching on alcohol content. so it's it's looking for undercarriage letters for all the words where needs to have normalized matching instead of exact matching. another issue is there isn't any weed outs blatantly telling the user exactly why a label has failed or passed the test and so we need to add that line within our ui. I added for this test I added 19 labels and overall the test took 11.88 seconds to complete with the longest individual individual label taking 1.4 seconds.

So we've overcome the OCR issue. out of our six clean labels now five of them past detection. within the five second

We came in the point of development today where OCR is now the weak point of our application. our application is running you can put in a CSV file a bunch of images and the application will extract the text from those images try to match each image to a particular record and CSV file then determine if there's a match needs review for so on and so forth. we came to a point in our development where we need to strengthen our OCR model because the text extracted a lot of the time is 15% of the time is unlegible or Not Human readable in short googly guy. Chad had to where I had to keep chat on in line because Chad wanted to update the the rules regarding the warning label and validation which at this point they need to stay straight so that we don't allow true errors to pass through as positive it's at this point where we need to update strengthen four to five or OCR model so that we're getting a higher percentage of breathable text once that is done then we can test to see if we need to change our rule set. We need to keep in mind Why OCR accuracy matters it ensures data Integrity improve searchability enables workflow automation helps with regulatory compliance.

So we started upgrading the OCR module. the updates took Took place in the ocr_service.pyscript. we first adapted the model to include grayscale conversion upscale image adapter threshold denoising dark brown dark background handling inverted pass. this made no recognizable changes in fact it seemed the results got worse less valid labels passed. first the first version of the script we got two out of the six to pass. on the second version of the script one out of six past. so the next upgrade was to add multiple passes of of a label using the original version a dark scale version Essentially we were doing eight image variants * 4 PSM modules which essentially gave us 32 runs per image this version of the file version 2 of the ocr_service.py file gave us 100% accuracy on the labels each label would pass with match consistency of 89% or greater however the processing time was well over the 5-minute required 5 Second requirement this is the tequila label took over 52 seconds 1 2 3 4 Brands to 16 seconds so on and so forth so the next attempt is to create a version of the script where we do have multiple passes however the one will stop once we get a legible pass. So now our process now we're going to try to attempt the OCR service will first try the fast OCR first score it stop early if it's good enough only use expensive passes when needed we will see how this one goes. In short or first version was cheap as in processing time and power however it was highly inaccurate. version 2 of the upgrade is highly accurate but it is very slow it's highly expensive it takes a lot of computing power and a lot of time.

After seven rounds of interracing we were able to get the majority of the valid records validated. what I mean by majority is five out of six records pass validation. we use the combination of having a maximum of six passes with the rule said that if all Fields were detected and validated within a pass then that label will exit the path logic or if a label does not is not validated within the past then it's passed on to the second or third or the next pass in the sequence. the other variable or value that we have is that passing the passing process will end if one label takes more than 5 seconds. after making these changes we had a result were five out of six labels are passing with the overall status pass however I noticed a flaw where two out of the five records

that pass match status is needs review my issue with this is that if a record has all Fields detected status is passed and the confidence scores 100%, then therefore these fields need to match up to a record in the CSV file. but with regards to orc extracting the data we've used we've mixed a bunch of different techniques for having six different passes happening where each will if so what happens is a label goes into the first pass information is extracted if that information feels it matches up with the field within that record whether it be brand name Class Type alcohol content net context government warning then that information is saved any if and if all fields are verified within that pass then the label leaves that pass. if all Fields aren't identified in the past then we keep the information from the field to detect it and the label moves on to the next pass and in that pass the results are focusing on the fields that we haven't identified yet pass. so this process continues to happen and to all Fields have been detected once all Fields have been detected and they're still passes left in the sequence the label exit the path logic anyway cuz we need no more information. the other variable is if after 5 Seconds all Fields have not been identified and verified the label X's the passing process anyway. so the next issue we need to overcome is the matching issue for records that have an overall pass status. secondly while doing this Development I've had chat update to UI to include a progress bar during the vallification process so that we know that the process is working and how long it's taking roughly for each label. so now we get what label the process is working on and a progress bar which increases or advances whenever a label has completed the process we've also add basic CSV exporting. however the issue with the CSV file is that once you click on one of the buttons whether it's Whether it's the download batch summary CSV or the download detailed validation CSV whenever that button is clicked either button is clicked the results for the batches reset. meaning that all all the detailed responses for each label is gone and to get to see that information again you would have to revalidate those labels. the other update I want to add to the UI is to add a seconds counter along with the progress bar so not only you know which image file is being processed but also how many seconds is taking for that label to be processed. overall overall status of the project so we have the UI up and running the buttons work the processing Works matching works for the most part we need to tweak that orc is better in strengthened I don't need I don't think we need to make any more changes to orc. rudimentary CSV downloading works we just strengthen we just need to strengthen the details. you need it I would like to add a second timer to the progress bar. I would like to strengthen the CSV download buttons whereas when you click the button it does not reset the data. and we need to add a fix to the matching status logic.

We've overcome the OCR issue out of our six clean labels five of them pass detection 61 is the majority of the fields however it cannot find the warning label within 5 Seconds we used a model where we have six different passes and each pass takes the out of after each pass whatever feels have been identified we keep that information so every other seceding pass only has to figure out what is left there's also a variable or if more than 5 Seconds have have gone by then we cut off running the pass. the issue we ran into this is we ran into after completing this We ran into another issue where after a label had considered to be passed with the conference score of 100 overall status pass all Fields detected with the status of pass and a confidence of 100 that match status was still set to review. so we had an update our logic so that after if if you if it wasn't matched to a record at the beginning we would rewind the matching sequence if all the

fields were complete so if we have all the fields and it passed and we're 100% confident therefore we can confirm that it is a match to a record in the CSV so we updated that so so that the results could be automatch confirmed or needs reviewed or also fail. we are now finished up the The project but adding a few Creature Comforts I would like to have a timer on the progress bar for how long each image is being scanned also the issue of when you click on a CSV file it clears out the data results so I would like to have that when you click the export CSV the results still stay there and that the only clear out whenever you run a new scan. once we do that I'm a stress test the application a little while with our valid poor quality and invalid labels to see how the system works. we will then look at our documentation and read me and make sure it fulfills all requirements and discuss all obstacles that we overcome. from there we will deploy this app so that it is available from the internet and then get ready to submit this for me.

All right so finally was able to get the update for app.pyscript complete. so now it will it can automatch confirm or needs review for particular record so we added a second pass to verify the matching after all if a record was not automation beginning and we found out that all Fields were detected and confirmed then we confirm that that record is a match to a record in our CSV file. from there we moved to the deploying our app to render.com or use the free tier. we first tried setting it up using the python pipeline however we couldn't get Tesseract OCR to work. so we ended up creating a Docker file within a project push that to the GitHub created a new project and render use the docker platform and that allowed OCR to work. I learned that Docker is a Linux container so we were able to run our Linux commands on python commands to that similar to how we do locally. I tried the verification test with our six labels and our Master CSV and the results were very slow Slow as in the longest process took 89.7 seconds and the shortest process to 27.3 seconds. the next step is to increase our compute power on render we're on the free version which we have I think 1/10 of a CPU and 1 GB of 1 MB of ram. I bumped it up to the standard plan which is \$25 a month and then we get an entire CPU and 2 MB of RAM. redeployed our application on renderer and once that is complete I will run the test again. Correction the amount of ram we have on renderer is 2 gigabytes.

Finally found the fix upgrading the plan helped a bit we went from 52 seconds to roughly like 20 seconds or so that was still way too high for the requirements end up chat end up changing the OCR service script we changed he changed it from 11 processes to eight still running slow after doing some research on how to make OCR through web applications one quicker one of the ideas was using multi-threading we implemented multi-threading in our code using os.environ and set it to one thread this alone brought down processing a Computing times drastically low from there I was playing with the max seconds and Max seconds per Permax Pass Time Out I was tweaking it it was supposed to be around roughly 5 seconds I had kicked it up to 10 then 15 seconds doing the 15 second interval that's when we implemented the OS single thread limiter line of code and processes dropped under the 5 second time however since we were using eight processes now processes were fending within like 3 and 1/2 seconds under the 5 second time frame however it was still missing fields we know that in our golden script the script that works the best you get five completely passed and one that needs review when we first implemented the changes we got one that passed in five that needs review halfway through implementing the max timer and the max pass timer along with the single thread we were

getting three past three with you there was another change there was an issue where within the new update the passes were being having a shared time similar to a cipher on BET and that's not what I wanted I wanted a Max time for total process for a particular label and then I wanted that label to pass through each process that process will generate all the taxi kid it would pass it on to the next it would fill out the fields needed then if all Fields weren't found it would pass the label off to the next process if there were time left and that process would try to figure out the remaining Fields once that process was done and all Fields weren't detected and it was still timeless it will continue the process on to the next process so on and so forth until the all Fields were detected time to ran out or all processes had completed however a processes weren't enough so we implemented back to 11 thankfully I had saved the earlier version of The Script around 6:35 p.m. this evening and so I rolled back to that one and I just implemented the OS line for single thread usage and that brought the script cuz that script originally was given results at 50 seconds and 38 seconds and so on . and that brought the process down from 50 seconds per label down to under 5 seconds per label so the fix was threading now everything is clear and updated the project is pretty much done at this point what I need to do is document read through the documents the read me updated add a section of what we what we found or we overcome how we got it working so on and so forth.